We thank all five reviewers for their careful reading and are encouraged that they find the work worthwhile — "sensible" (27Jx), "important"/"promising" (D8hx), and "a fair and interesting contribution" with "compelling" matched-setting results (wGgF). We take the presentation problems seriously and have completed a full language and formatting revision, verifying that no model, experiment, or reported result is affected by these edits.

Reviewer 27Jx (#1)—R1-1: Presentation. We sincerely apologize for the draft-stage errors, and we are very grateful for the reviewer's careful reading. We have carried out a thorough revision: resolved the dangling "Figure ??" to Fig. 1 (Sec. 3.2), corrected the flagged typos ("multi-granul"→"multi-granular", "Constrion"→"Construction", "Fame-level"→"Frame-level"), re-rendered Figs. 1–2 as vector graphics, and made a careful, line-by-line language pass over the entire manuscript. These were purely presentation issues, not substantive ones; no result, number, or claim was affected.

R1-2: Distinction from Vgent. We thank the reviewer. The distinction is paradigmatic and does not hinge on any single layer. Vgent is a *fixed* retrieve-then-read graph-RAG over frame and segment local relations. In contrast, our contribution is twofold: we build a multi-level scene-graph index over long video, and we design a modular, plug-and-play adaptive agent that uses the scene graph as its primary retrieval index, verifying each proposal against raw frames before answering (R2-3). Together they drive the +18.2-pp matched-setting gain on LVBench (54.8%→73.0%). We address the event layer's specific role, and its accuracy contribution, in R1-3.

R1-3: The role of the event layer. We agree with the observation. At a fixed backbone the event layer's measurable effect is efficiency: it cuts LLM and frame-inspection calls by 71.5% and 74.7% at unchanged accuracy (64.6%→64.4%). We read this as a capability gain, not a void: reaching the same answers with far less search is itself better reasoning. Still, efficiency need not carry the contribution alone. The flat accuracy reflects the conservative event-graph construction of the submitted version, not an intrinsic ceiling. In an additional experiment that strengthens the construction and queries the index *caption-free*, the agent reaches **68.5%** e2e accuracy on LVBench, a 3.9-pp gain over the segment-only version and a 13.7-pp gain over the caption-only baseline. A graph that answers 68.5% of questions with *no* captions is an answer-bearing reasoning substrate, so the distinction does *not* collapse. The same structured index lets the lighter `o4-mini` reach 73.0%, matching DVD's far stronger `o3` (74.2%) and beating DVD's own `o4-mini` (70.2%). Our continued experiments improve these results further; we will report fair, matched comparisons in the camera-ready.

Reviewer D8hx (#2)—R2-1: Offline construction cost. For all 103 LVBench videos (115.5h), construction costs $112.74 (12,360 `gpt-4.1` frame-graph calls at one keyframe per 5s, ≤120/video; 7.94h on four workers; the symbolic Stage 3 event build adds only 121s), i.e. $0.98 per video-hour. We reuse DVD's open-sourced captions rather than generating our own, so this is our only added offline cost. End-to-end VQA over all 1549 questions costs $279.54 (`o4-mini`) and $499.31 (`o3`), so this one-time construction is only 40% and 23% of the recurring online budget it serves.

R2-2: Cross-benchmark control. We agree not all entries are controlled. On LVBench the DVD comparison *is* fully controlled: the same `o3` backbone, DVD's own open-sourced captions, and the same video database. On the other three, DVD released no full reproduction setup, so we cannot match its pipeline; we still use its captions and report these as best-effort positions. Other commercial and open models have closed weights, so they are reference positions only. Our claims therefore rest on the fully controlled LVBench comparison and its matched-setting ablations.

R2-3: Direct graph quality. Direct evaluation is desirable, but intrinsic scene-graph metrics need a gold-annotated benchmark that does not exist at hour scale, and they would still miss what matters: extraction accuracy and the graph's net effect on answers. Comparable systems, DVD included, are judged the same way, not by intrinsic quality. We therefore measure the graph through the task: the caption-free graph-only setting is cleanest, since with captions removed the answer must come from the graph, yet accuracy holds at 68.5% on LVBench (R1-3), and the matched ablations isolate its gain. We will still add a small human-verified sample for entity and relation precision; the pipeline already shows 76.8% entity-mention absorption and a 5.0% invalid-relation rate. Propose–verify re-inspection of the raw frames further reduces how much frame-level error propagates into the event graph.

R2-4: Implementation details. To give the values asked for: entity alignment scores name, description, and type similarity with weights 0.7, 0.2, and 0.1, and merges above a threshold of 0.78; the entity blacklist removes artefacts and over-generic terms; segments merge into events at a 12s temporal gap with relation-overlap thresholds. The composite retrieval score and all remaining thresholds are fully specified in the appendix, which gives the alignment, gating, and event-construction algorithms; we will release a fully reproducible repository for the camera-ready.

R2-5: Presentation. See R1-1.

R2-6: Isolating localization from FTT. Our reported answered accuracy (ans_acc) separates the two: the caption-only baseline scores 65.7% and SGDAgent 74.2% (+8.5 pp), independent of the answer rate, while the FTT drop (16.5%→1.6%) is itself a consequence of structured retrieval converging where open caption search loops for lack of locatable evidence.

Reviewer k56s (#3)—R3-1: Baseline data citation. Thank you for catching this. The correct DVD values are Video-MME 67.3 (we had mis-copied DVD's LongVideoBench(Long) entry, 68.6) and EgoSchema 76.6, both matching DVD's published numbers. The correction in fact widens our Video-MME margin to 69.0 vs. 67.3. We will present a carefully re-verified version, with a more general Fig. 1 example, in the camera-ready.

R3-2: Related works. Thank you; we will cite and discuss all three ([1] Mind Palace, [2] the Gaussian-token work, [3] GraphVideoAgent) in the camera-ready. They use the graph as an in-context representation or online entity memory, whereas we use it as the agent's primary retrieval index, gated to raw-frame verification before answering (see R1-2), a distinct design.

R3-3: Offline construction cost. See R2-1.

R3-4: Error propagation & robustness. Two mechanisms limit it. First, propose–verify gates every graph hit to raw-frame re-inspection before answering (R2-3), so a hallucinated or mis-aligned node is caught at verification rather than read out as an answer, reducing upward propagation. Second, the caption-free graph-only system still reaches 68.5% (R1-3), evidence that the pipeline tolerates imperfect frame-level extraction. To quantify this we will add a robustness analysis that perturbs frame-graph quality and reports the accuracy sensitivity, and we will discuss [4].

Reviewer SSkz (#4)—R4-1: Over-strong coverage claim. We agree, and will revise the wording: the method improves *localization within the sampled frames*, and we will no longer claim to address incomplete sampling coverage, since the graph itself is built only from sparsely sampled keyframes.

R4-2: The role of the event graph. In the submitted table the event graph is not a fixed-backbone accuracy driver, and we present it as a higher-order layer that reduces the agent's search burden. A strengthened, caption-free run, however, shows it also carries accuracy, not only efficiency (see R1-3).

R4-3: Heuristic construction. For robustness see R2-3; we now fully specify the alignment, merging, and event thresholds (see R2-4).

R4-4: Presentation. See R1-1; we have already re-rendered Figures.

R4-5: Questions. Scene-graph accuracy: see R2-3. Repositioning the event graph as an efficiency layer: it does reduce search burden, but a strengthened caption-free run shows it is also accuracy-bearing (see R1-3). Full pipeline cost per video: see R2-1. Sampling interval: we currently sample one frame every 5s and will report 2s/5s/10s sensitivity in the revision.

Reviewer wGgF (#5)—R5-1: HIG & hierarchy novelty. Thank you; we agree HIG proposes a hierarchical video graph, and we will cite and contrast it. We will not claim the hierarchy concept as novel. Our work differs in task: HIG targets video scene-graph *generation*, whereas we use a graph index to *drive an adaptive agent* for long-video QA (see R1-2).

R5-2: Implementation details. The alignment, retrieval, gating, and event-construction rules are given in the appendix as pseudocode; see R2-4 for the headline values.

R5-3: Graph-depth ablation. See R1-3: the segment graph drives fixed-backbone accuracy; the event graph reduces search burden and, under a stronger construction, also contributes accuracy (caption-free 68.5%).

R5-4: Efficiency evidence. Beyond online call counts we now also report dollar cost and construction time (see R2-1).

R5-5: Presentation. See R1-1.